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| University of Bristol - EMATM0051 (LSDE) Coursework |
| WordFreq App - Overview and Setup Instructions |

|Original App (c) AWS Inc. |

Overview

Word Frequency is a sample service built with AWS SDK for Go.
The service takes advantage of Amazon EC2 service, Amazon Simple Storage service, Amazon Simple Queue Service and Amazon DynamoDB to collect and report the top 10 most common words of a text file.
The original sample app released by AWS is here:
https://github.com/aws-samples/aws-go-wordfreq-sample

Setup guide

The rest of this document covers step by step setup of the standard application, ready for further work required in the LSDE coursework (see the separate coursework documentation).

Pre-requisites

- AWS Academy Foundation Learner Lab account, registered and activated
- SSH Terminal / PuTTy or similar on Windows
- Internet Browser - Google Chrome or Firefox are best

Task A - Launching the Development Instance

1. Log in to AWS Academy and start up the Learner Lab
2. On the lab screen note the remaining credits, and check this as you start each session.
   -- $40 should be more than enough for the coursework and all testing
   -- If you are down to $10 or under shut down any EC2 instances and contact the instructor for advice.
3. Note the session time. You have 4 hours before you need to refresh this Lab page and reopen any AWS Console pages.
   -- All resources you create will remain in your Learner Lab for as long as you have credit available, they are not destroyed at the end of a session!
4. Click the 'AWS' button with green status once the Lab has started, to open a new AWS Console page.
5. Go to the EC2 service and launch an instance with the following non-default settings:
   -- AMI:
   --- Select Ubuntu
   --- Select the default AMI (Ubuntu Server 22.04 LTS(HVM), SSD Volume Type)
   -- Name: wordfreq-dev
   -- Instance Type: t2.micro
   -- KeyPair: name: learnerlab-keypair (create a new keypair then download the .pem file and keep it safe! If using PuTTy on Windows, please download the .ppk file)
   -- Security Group: allow SSH access (default rule)

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|布里斯托大学 - EMATM0051 (LSDE) 课程 |

| WordFreq 应用程序 - 概述和设置说明 |

|原始应用程序 (c) AWS Inc. |

概述

WordFrequency 是使用 AWS SDK for Go 构建的示例服务。

该服务利用 Amazon EC2 服务、Amazon Simple Storage 服务、Amazon Simple Queue Service 和 Amazon DynamoDB 来收集和报告文本文件中最常见的 10 个单词。

AWS 发布的原始示例应用程序位于：https://github.com/aws-samples/aws-go-wordfreq-sample

设置指南

本文档的其余部分涵盖了标准应用程序的逐步设置，为 LSDE 课程作业中所需的进一步工作做好准备（请参阅单独的课程作业文档）。

先决条件

- AWS Academy Foundation Learner Lab 帐户，在 Windows Internet 浏览器上注册并激活 SSH 终端/PuTTy
- 或类似帐户 - Google Chrome 或 Firefox 最好
-

任务 A - 启动开发实例

1.登录AWS Academy并启动学习者实验室

2. 在实验屏幕上记下剩余学分，并在开始每个课程时检查这一点。

   -- 40 美元对于课程作业和所有测试来说应该绰绰有余

   -- 如果您的费用降至 10 美元或以下，请关闭所有 EC2 实例并联系讲师寻求建议。

3. 注意会话时间。您有 4 小时的时间需要刷新此实验室页面并重新打开任何 AWS 控制台页面。

-- 只要您有可用积分，您创建的所有资源都将保留在您的学习者实验室中，它们不会在课程结束时被销毁！

4. 实验开始后，单击绿色状态的“AWS”按钮，打开新的 AWS 控制台页面。

5. 转到 EC2 服务并启动具有以下非默认设置的实例：

   -- AMI：

   --- 选择Ubuntu

   --- 选择默认的 AMI（Ubuntu Server 22.04 LTS(HVM)，SSD 卷类型）

   -- 名称：wordfreq-dev

   -- 实例类型：t2.micro

   -- 密钥对：名称：learnerlab-keypair（创建新密钥对，然后下载 .pem 文件并妥善保管！如果在 Windows 上使用 PuTTy，请下载 .ppk 文件） -- 安全组：允许 SSH 访问（默认规则）

-- Storage Volume: 30GB GP2
-- IAM instance profile (under Advanced Details): choose EMR_EC2_DefaultRole

Task B - Create the S3 Buckets

1. Select S3 using the Services dropdown at the top left of the EC2 Console page (it can be helpful to open this in a new browser tab).
2. Create a new S3 Bucket with the following non-default settings. This is your uploading bucket:
   -- Bucket name: a unique name, using alphanumeric characters or dashes, perhaps using your initials or date;
   e.g. zj-wordfreq-nov24-uploading lyc-wordfreq-nov24-uploading
3. Create the second S3 Bucket with the following non-default settings. This is your processing bucket:
   -- Bucket name: a unique name, using alphanumeric characters or dashes, perhaps using your initials or date;
   e.g. zj-wordfreq-nov24-processing lyc-wordfreq-nov24-processing
4. Make a note of your bucket names for later.
5. Note the bucket ARNs (Amazon Resource Names): on the Bucket list view, select a queue and click 'Copy ARN', then note each bucket ARN for later.

Note: The uploading S3 Bucket is used for uploading text files from your local machine and storing those files.
The files will be copied to your processing S3 bucket.
The bucket will have upload notifications enabled, such that when a file is uploaded, a message notification is automatically added to a wordfreq SQS queue.

Task C - Create the SQS Queues

1. Select SQS using the Services dropdown, ideally opening in another new browser tab.
2. Create a new SQS queue (for file processing jobs) with the following non-default settings:
   -- Queue type: Standard
   -- Queue name: wordfreq-jobs wordfreq-jobs
   -- Access policy: Advanced
   -- Change the JSON policy code section that looks like this...:

```
"Principal": {
    "AWS": "<12 digits>"
  },
```

-- ...to the following (this allows any AWS entity to write to the queue, not just the queue owner):

```
"Principal": {
    "AWS": "*"
  },
```

3. Once the queue is created, take a note of the queue URL ('https://sqs.us-east-1.amazonaws.com/....').
4. Create a second SQS queue (for file processing results) with the following non-default settings:

-- 存储卷：30GB GP2 -- IAM 实例配置文件（在“高级详细信息”下）：选择 EMR_EC2_DefaultRole

任务 B - 创建 S3 存储桶

1. 使用 EC2 控制台页面左上角的“服务”下拉列表选择 S3（在新的浏览器选项卡中打开它会很有帮助）。

2. 使用以下非默认设置创建新的 S3 存储桶。这是您的上传存储桶：

-- 存储桶名称：唯一的名称，使用字母数字字符或破折号，也可以使用您的姓名缩写或日期；例如zj-wordfreq-nov24-uploading
3. 使用以下非默认设置创建第二个 S3 存储桶。这是您的处理存储桶：
-- 存储桶名称：唯一的名称，

使用字母数字字符或破折号，可能使用您的姓名首字母或日期；例如zj-wordfreq-nov24-处理

4. 记下您的存储桶名称以供稍后使用。
5. 记下存储桶 ARN（Amazon 资源名称）：在存储桶列表视图中，选择一个队列并单击“复制 ARN”，然后记下每个存储桶 ARN 以供稍后使用。

注意：上传S3 Bucket用于从本地计算机上传文本文件并存储这些文件。

文件将被复制到您的处理 S3 存储桶。
该存储桶将启用上传通知，这样当上传文件时，消息通知会自动添加到 wordfreq SQS 队列中。

任务 C - 创建 SQS 队列

1. 使用服务下拉列表选择 SQS，最好在另一个新的浏览器选项卡中打开。
2. 使用以下非默认设置创建新的 SQS 队列（用于文件处理作业）：
   -- 队列类型：标准
   -- 队列名称：wordfreq-jobs
   -- 访问策略：高级
   -- 更改 JSON 策略代码部分，如下所示...：

```
“主要的”：｛
   "AWS": "<12 位>" },
```

-- ...以下内容（这允许任何 AWS 实体写入队列，而不仅仅是队列所有者）：

```
“主要的”：｛
    “AWS”：“*”},
```

3. 创建队列后，记下队列 URL ('https://sqs.us-east-1.amazonaws.com/')。

4. 使用以下非默认设置创建第二个 SQS 队列（用于文件处理结果）：

-- Queue name: wordfreq-results wordfreq-results
-- Access policy: Advanced [configure as for the jobs queue]

5. Once again, make a note of the queue URL once it's created.

https://sqs.us-east-1.amazonaws.com/381492240521/wordfreq-jobs
https://sqs.us-east-1.amazonaws.com/381492240521/wordfreq-results

Task D - Configure the File Copied notification from Bucket to SQS

1. Return to the S3 Console page and click on your processing S3 Bucket (e.g. zj-wordfreq-nov23-processing) > Properties.
2. Scroll down to 'Event notifications', click 'Create event notification'.
3. Configure the following non-default settings:
   -- Event name: file-copied-ev file-copied-ev
   -- Event types: [Select 'All object create events']
   -- Destination: SQS queue
   -- SQS queue: [Select 'wordfreq-jobs']

Task E - Log in to the Dev Instance and Setup the Environments

1. Return to the EC2 Console and select the wordfreq-dev instance (select the checkbox).
2. Click the Connect button above and select the 'SSH client' tab.
3. The connection instructions are correct if your PC is running Linux or MacOS (in a Terminal window):
   -- If you are connecting from a Windows PC, following instructions in sections 'Prerequisites' and 'Connecting to your Linux instance' on the following page:
   https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/putty.html
   (This is for PuTTy - other SSH clients are available)
   There is also a helpful 4-min video available from Linux Academy (ensure you use the 'ubuntu' username when logging in):
   https://www.youtube.com/watch?v=bi7ow5NGC-U&lc=Ugh4DAc2SqJj3HgCoAEC

   NOTE: If using PuTTy, pasting text from the system clipboard onto the command line is often achieved using a mouse right-click only.
4. When logging in for the first time, you may need to confirm the connection is valid by typing 'yes'.
5. You should see a command line prompt of the form: 'ubuntu@ip-172-XXX-XXX-XXX' OR ':~ $' (it may take 30 seconds to finally display) - this confirms you have logged in successfully.
6. In your SSH CLI window and run the following command to update the system (it may take a minute or two):

sudo apt update

7. Install AWS CLI and unzip

sudo snap install aws-cli --classic
sudo apt install unzip

8. Install and Configure go language

-- 队列名称：wordfreq-results
-- 访问策略：高级 [配置作业队列]
5. 创建队列 URL 后，再次记下该队列 URL。

任务 D - 配置从存储桶到 SQS 的文件已复制通知

1. 返回 S3 控制台页面，然后单击您的处理 S3 存储桶（例如 zj-wordfreqnov23-processing）> 属性。

2. 向下滚动到“事件通知”，单击“创建事件通知”。
3. 配置以下非默认设置：
   -- 事件名称：file-copied-ev
   -- 事件类型：[选择“所有对象创建事件”]
   -- 目的地：SQS队列
   -- SQS 队列: [选择 'wordfreq-jobs']

任务 E - 登录到开发实例并设置环境

1. 返回 EC2 控制台并选择 wordfreq-dev 实例（选中复选框）。

2. 单击上面的“连接”按钮并选择“SSH 客户端”选项卡。

3. 如果您的 PC 运行的是 Linux 或 MacOS（在终端窗口中），则连接说明是正确的：
-- 如果您从 Windows PC 连接，

以及下一页上的“连接到您的 Linux 实例”：
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/putty.html（这是针对 PuTTy - 其他
SSH 客户端可用）还有一个Linux Academy 提供了有用的 4 分钟视频（确保您使用

登录时的“ubuntu”用户名）：https://www.youtube.com/watch?v=bi7ow5NGC-U&lc=Ugh4DAc2SqJj3HgCoAEC

注意：如果使用 PuTTy，通常只需单击鼠标右键即可将系统剪贴板中的文本粘贴到命令行上。

4. 首次登录时，您可能需要输入“yes”来确认连接是否有效。

5. 您应该看到以下形式的命令行提示符：'ubuntu@ip-172-XXX-XXX-XXX' OR ':~ $'（可能需要 30 秒才能最终显示）
- 这确认您已成功登录。

6. 在 SSH CLI 窗口中运行以下命令来更新系统（可能需要一两分钟）：

须藤apt更新

7.安装AWS CLI并解压

sudo snap install aws-cli --经典 sudo apt install unzip

8.安装并配置go语言

8a. Download the package:
  wget https://go.dev/dl/go1.20.1.linux-amd64.tar.gz
8b. Unzip the file and move it to the directory
  sudo tar -C /usr/local -xzf go1.20.1.linux-amd64.tar.gz
8c. Add to the $PATH environment variable:
  vi ~/.bash_profile
-- Press 'i' to enter Insert mode in Vi, then paste or type the following:
  export PATH=$PATH:/usr/local/go/bin
-- Now exit Insert mode by pressing the Escape key (Esc)
-- Enter the following key strokes to write updates and quit: colon, lower case 'w' and lower case 'q', then press ENTER, i.e.

:wq

8d. Reload the profile and Check if Go is installed and its version:
  source ~/.bash_profile
  go version

If you see the similar output 'go version go1.20.1 linux/amd64', which indicates that go has successfully been installed.

NOTE: To quit the SSH session later, type: exit

Task F - Copy the Application Code zip onto the Dev Instance

1. Return to the S3 Console page and click on your uploading Bucket name, e.g. zj-wordfreq-nov24-uploading
2. Click 'Upload' on the 'Objects' tab and select the coursework zip file ('lsde-wordfreq-app.zip'), OR drag the coursework zip file directly onto this webpage and confirm.
3. Once uploaded, click on the blue zip filename link now present in the 'Files and Folders' section.
4. On the file details page that opens, find the S3 URI and click the copy button next to it (or click the 'Copy S3 URI' button) and note it down - we will use this to access the file from the CLI later;
   e.g. s3://zj-wordfreq-nov24-uploading/lsde-wordfreq-app.zip
5. Type the following 'S3 list' command to ensure you can see your S3 Bucket name, which shows you have correct permissions. You should be able to see two S3 buckets

aws s3 ls

6. Run the following command, entering your noted down 'S3 URI' instead of S3_URI (don't forget the final dot, with a space before it, indicating to copy to the current directory):

aws s3 cp S3_URI .

7. Check you now have the zip file downloaded on your Dev Instance, then unzip the package:

ls
unzip lsde-wordfreq-app.zip

8. Run the 'ls' command again - you should now see you have a new 'lsde-wordfreq-app' directory.

8a.下载包：
wget https://go.dev/dl/go1.20.1.linux-amd64.tar.gz
8b.解压文件并将其移动到目录
sudo tar -C /usr/local -xzf go1.20.1.linux-amd64.tar.gz
8c.添加到 $PATH 环境变量：
vi ~/.bash_profile
-- 在 Vi 中按“i”进入插入模式，

然后粘贴或键入以下内容： export PATH=$PATH:/usr/local/go/bin -- 现在按 Escape 键 (Esc) 退出插入模式
-- 输入以下按键写入更新并退出：冒号、小写“w”和小写“q”，然后按 ENTER，即

:wq

8d.重新加载配置文件并检查 Go 是否已安装及其版本：

source ~/.bash_profile

go version

如果看到类似的输出‘go version go1.20.1 linux/amd64’，这表明go已经

已成功安装。

注意：要稍后退出 SSH 会话，请键入：exit

任务 F - 将应用程序代码 zip 复制到开发实例上

1. 返回S3控制台页面，点击您的上传Bucket名称，例如： zj-wordfreqnov24-uploading 2. 单击“对象”选项卡上的“上传”，然后选择课程作业 zip 文件（“lsde-wordfreq-app.zip”），或将课程作业 zip 文件直接拖到此网页上并确认。

3. 上传后，单击“文件和文件夹”部分中现在显示的蓝色 zip 文件名链接。 4. 在打开的文件详细信息页面上，找到 S3 URI，然后单击其旁边的复制按钮（或单击“复制 S3 URI”按钮）并记下它 - 稍后我们将使用它从 CLI 访问文件;例如s3://zj-wordfreq-nov24-uploading/lsde-wordfreq-app.zip 5. 键入以下“S3 list”命令以确保您可以看到您的 S3 存储桶名称，这表明您拥有正确的权限。您应该能够看到两个 S3 存储桶

aws s3 ls

6. 运行以下命令，输入记下的“S3 URI”而不是 S3_URI（不要忘记最后一个点，前面有一个空格，表示复制到当前目录）：

aws s3 cp S3_URI 。

7. 检查您现在是否已将 zip 文件下载到您的开发实例上，然后解压缩该包：

ls 解压 lsde-wordfreq-app.zip

8. 再次运行“ls”命令 - 您现在应该看到一个新的“lsde-wordfreq-app”目录。

NOTE: We will leave the zip file in case you need to completely delete the directory AND all its files and start again.
You can do this with the following command in the home directory: rm -rf lsde-wordfreq-app; unzip lsde-wordfreq-app.zip

Task G - Set up and Configure the WordFreq App

1. Change directory to the app folder and ensure all shell scripts have correct execution permissions:

cd lsde-wordfreq-app
chmod +x *.sh

2. Run the 'setup.sh' script, which will install the GO language runtime and any dependencies, as well as creating the DynamoDB 'wordfreq' database table:

./setup.sh

NOTE: This script should take a couple of minutes to run, and end with 'Setup complete.'. If errors are shown, run it again.
If there are still other errors (ignore the 'table already exists' error), and you don't get 'Setup complete', please add a post on the BB forum or book an Office Hours session.

3. Optional: In another browser tab, open the DynamoDB Console, click Tables, select wordfreq and click 'Actions', then 'Explore Items', which will display items (rows) added to the table. Initially the table is empty.
4. You will now need to manually edit the 'run_worker.sh' scripts to refer to the correct SQS queue URLs.
-- These instructions assume you will use the 'Vi' editor, but you can install and use any other one, such as GEdit or EMACS, as required.
-- Type the following to open and edit the run_worker.sh script in Vi:

vi run_worker.sh

-- Using the arrow keys, move the cursor down to the following line:

export WORKER_QUEUE_URL="https://sqs.us-east-1.amazonaws.com/12345678/wordfreq-jobs"

-- Press 'i' to enter Insert mode in Vi, then delete the URL and paste or type in your noted URL for the SQS jobs queue between the quotes.
-- Similarly edit the following line, updating the URL value for the results queue with your own:

export WORKER_RESULT_QUEUE_URL="https://sqs.us-east-1.amazonaws.com/12345678/wordfreq-results"

-- Now exit Insert mode by pressing the Escape key (Esc)
-- Enter the following key strokes to write updates and quit: colon, lower case 'w' and lower case 'q', then press ENTER, i.e.

:wq

Task H - Test the Worker and Upload functionality

1. We will now try to test the basic app functionality. We first need to empty the jobs queue of any spurious messages.

注意：我们将保留 zip 文件，以防您需要完全删除目录及其所有文件并重新开始。

您可以在主目录中使用以下命令来执行此操作：rm -rf lsde-wordfreq-app;解压 lsde-wordfreq-app.zip

任务 G - 设置和配置 WordFreq 应用程序

1. 将目录更改为 app 文件夹并确保所有 shell 脚本具有正确的执行权限：

cd lsde-wordfreq-app

chmod +x *.sh

2. 运行“setup.sh”脚本，该脚本将安装 GO 语言运行时和任何依赖项，并创建 DynamoDB“wordfreq”数据库表：

./setup.sh

注意：此脚本应该需要几分钟才能运行，并以“安装完成”结束。如果显示错误，请再次运行。

如果仍然存在其他错误（忽略“表已存在”错误），并且您未收到“设置完成”消息，
请在 BB 论坛上添加帖子或预订“办公时间”会议。

3. 可选：在另一个浏览器选项卡中，打开 DynamoDB 控制台，单击表，选择 wordfreq 并单击“操作”，
然后单击“浏览项目”，这将显示添加到表中的项目（行）。

   最初表是空的。

4. 现在，您需要手动编辑“run_worker.sh”脚本以引用正确的 SQS 队列 URL。

   -- 这些说明假设您将使用“Vi”编辑器，但您可以根据需要安装和使用任何其他编辑器，例如 GEdit 或 EMACS。

   -- 键入以下内容以在 Vi 中打开并编辑 run_worker.sh 脚本：

vi run_worker.sh

-- 使用箭头键将光标向下移动到以下行：

导出 WORKER_QUEUE_URL="https://sqs.us-east-1.amazonaws.com/12345678/wordfreq-jobs"

-- 按“i”进入 Vi 中的插入模式，然后删除 URL 并在引号之间粘贴或键入您记下的 SQS 作业队列 URL。

-- 同样编辑以下行，用您自己的值更新结果队列的 URL 值：

导出 WORKER_RESULT_QUEUE_URL="https://sqs.us-east-1.amazonaws.com/12345678/wordfre q-results"

-- 现在按退出键 (Esc) 退出插入模式 -- 输入以下按键来写入更新并退出：冒号、小写“w”和小写“q”，然后按
ENTER，即

:wq

任务 H - 测试 Worker 和上传功能

1. 我们现在将尝试测试基本的应用程序功能。我们首先需要清空作业队列中的任何虚假消息。

-- Return to the SQS Console window and select the 'wordfreq-jobs' queue (click on the radio button on the left)
-- Click on the 'Actions' button, then the 'Purge' button and type in 'purge' where required to confirm you want to delete all messages.

2. SSH Window: Worker
   -- Ensure you are in the correct directory and start the worker process. If there are errors, check your SQS URLs in the run_worker.sh file.

cd ~/lsde-wordfreq-app
./run_worker.sh

-- You should see some lines of log output, which will increase when the worker finds jobs to process.

NOTE: The main WordFreq process is this 'worker' application, which runs continuously, checking for jobs on the queue and processing them.

3. Upload one of the text files from data.zip on BB to your to your 'processing' S3 bucket, e.g. zj-wordfreq-nov24-processing

-- You should see after a few moments that the worker retrieves the job and successfully processes the job. You could check the results in your DynamoDB table.

NOTE: The worker performance has been deliberately crippled by adding an extra 'wait' of between 10-20 seconds during processing, which you must not modify.
This makes it much easier to ensure the scaling operation is effective without requiring hundreds of input files.

-- If you observed the output described above, the basic application is working. We now just need to set it up as a Linux service.

Task I - Setting up the Worker Service

1. In your SSH Window:
   -- Press CTRL+c to exit the worker.sh process.
   -- Set up the WordFreq Worker service by running the shell script:

./configure-service.sh

NOTE 1: This command installs the Worker.sh command as a service, which runs in the background and will auto-start on boot.
It's important that any auto-scaling EC2 worker instances have this service configured in this way.

NOTE 2: If you do NOT get a 'Service started successfully' message, run again, or run through the setup process again.
If you still experience issues, please post on the BlackBoard Discussion Forum for LSDE, or book an Office Hours session.

-- To view the output logs from the running wordfreq worker service, enter the following (CTRL+c to exit):

sudo journalctl -f -u wordfreq

NOTE: To stop or start the wordfreq worker service, run the following commands, respectively:

-- 返回到 SQS 控制台窗口并选择“wordfreq-jobs”队列（单击左侧的单选按钮）
-- 单击“操作”按钮，然后单击“清除”按钮并输入“清除”，其中需要确认您要删除所有消息。

2. SSH 窗口：Worker -- 确保您位于正确的目录中并启动工作进程。
如果存在错误，请检查 run_worker.sh 文件中的 SQS URL。

cd ~/lsde-wordfreq-app ./run_worker.sh

-- 您应该看到一些日志输出行，当工作人员找到要处理的作业时，这些行会增加。

注意：主要的 WordFreq 进程是这个“工作”应用程序，它连续运行，检查队列中的作业并处理它们。

3. 将 BB 上的 data.zip 中的文本文件之一上传到“处理”S3 存储桶，例如zjwordfreq-nov24-处理

-- 片刻之后，您应该会看到工作人员检索作业并成功处理作业。您可以在 DynamoDB 表中检查结果。

注意：通过添加额外的“等待”，故意削弱了工作人员的表现
处理期间 10-20 秒之间，您不得修改。
这使得更容易确保缩放操作有效，而不需要数百个输入文件。
-- 如果您观察到上述输出，则基本应用程序正在运行。我们现在只需将其设置为 Linux 服务即可。

任务 I - 设置 Worker 服务

1. 在 SSH 窗口中： -- 按 CTRL+c 退出worker.sh 进程。

-- 通过运行 shell 脚本来设置 WordFreq Worker 服务：

./configure-service.sh

注意 1：此命令将 Worker.sh 命令安装为服务，该服务在后台运行，并在启动时自动启动。

重要的是，任何自动缩放 EC2 工作线程实例都在此配置此服务
way.

注意 2：如果您没有收到“服务已成功启动”消息，请再次运行，或再次运行安装过程。

如果您仍然遇到问题，请在 LSDE 的 BlackBoard 讨论论坛上发帖，或预订办公时间会议。

-- 要查看正在运行的 wordfreq 辅助服务的输出日志，请输入以下内容 (CTRL+c
退出）：

须藤journalctl -f -u wordfreq

注意：要停止或启动 wordfreq Worker 服务，请分别运行以下命令：

sudo systemctl stop wordfreq
sudo systemctl start wordfreq

2. In your processing S3 bucket:
   -- We need to upload the test file again to check our new service processes it correctly
   -- Upload any text file (You could use any text file from data.zip on BB) to your to your 'processing' S3 bucket, e.g. zj-wordfreq-nov24-processing

3. Check again that SSH Window shows the worker output in the log entries.
4. At this point, press CTRL+c in your SSH Window if you don't need it anymore, and pat yourself on the back, we're done!
   -- BUT ... make an AMI backup of this EC2 instance - see 'strong recommendation' below - THEN you can relax. ;-)

Task J - Upload files to your S3 uploading bucket

We will use the two S3 buckets to simplify the uploading of files for processing by the application

1. Download data.zip from BB.
2. Unzip the downloaded file and review the files in the folder. You should have about 120 text files in the folder.
   Ensure:
   - Files are in text (.txt) format only, no binary files should be used.
   - Filenames have no spaces or non-alphanumeric characters (hypens, underscores are ok).
3. Upload these files to your 'uploading' S3 bucket, e.g. zj-wordfreq-nov24-uploading

Task K - Copy files to your S3 processing bucket to start processing them

1. To start processing these files with your worker application service, we simply copy the text files (only) from the uploading bucket to the processing bucket.
   -- NOTE: This operation may take around 20 minutes.
   -- Run the following S3 copy command in SSH

aws s3 cp s3:// s3:// --exclude "" --*include* ".txt" --recursive

e.g.
aws s3 cp s3://zj-wordfreq-nov24-uploading s3://zj-wordfreq-nov24-processing --exclude "" --*include* ".txt" --recursive

2. Observe the processing log as the application begins to process the files:
   sudo journalctl -f -u wordfreq

Task L - Consult the Coursework Doc for tasks

- When implementing and testing autoscaling, you will be mainly using the following operations of those we have learned here:
  -- Copying files from the uploading bucket to the processing bucket to start processing them (Task K)
  -- Purging (emptying) messages from the queues (Use the 'purge' button in the SQS console - Task H(1))
  -- Reviewing the worker logs on an EC2 instance (Task I/K)

运行以下命令 sudo systemctl stop wordfreq sudo systemctl start wordfreq

2. 在您的处理 S3 存储桶中： -- 我们需要再次上传测试文件，以检查我们的新服务是否正确处理它
-- 将任何文本文件（您可以使用 BB 上 data.zip 中的任何文本文件）上传到您的'处理' S3 存储桶，例如 zj-wordfreq-nov24-processing

3. 再次检查 SSH 窗口是否在日志条目中显示工作程序输出。
4. 此时，如果您不再需要它，请在 SSH 窗口中按 CTRL+c，然后拍拍自己的背，我们就完成了！
-- 但是...对此 EC2 实例进行 AMI 备份 - 请参阅下面的“强烈建议”， 然后您就可以放松了。 ;-)

任务 J - 将文件上传到您的 S3 上传存储桶

我们将使用两个 S3 存储桶来简化文件上传以供应用程序处理

1. 从BB下载data.zip。
2. 解压下载的文件并查看文件夹中的文件。该文件夹中应该有大约 120 个文本文件。

   确保：
   - 文件仅采用文本 (.txt) 格式，不应使用二进制文件。
   - 文件名没有空格或非字母数字字符（连字符、下划线都可以）。
3. 将这些文件上传到您的“上传”S3 存储桶，例如 zj-wordfreq-nov24-uploading

任务 K - 将文件复制到 S3 处理存储桶以开始处理它们

1. 要开始使用工作线程应用程序服务处理这些文件，我们只需将文本文件（仅）从上传存储桶复制到处理存储桶。

   -- 注意：此操作可能需要大约 20 分钟。

   -- 在 SSH 中运行以下 S3 复制命令

aws s3 cp s3:// s3:// --exclude "" --include ".txt" --recursive

e.g.
aws s3 cp s3://zj-wordfreq-nov24-uploading s3://zj-wordfreq-nov24-processing --exclude "" -include ".txt" --recursive

2. 当应用程序开始处理文件时观察处理日志： sudo Journalctl -f -u wordfreq

任务 L - 请查阅课程作业文档以了解任务

- 在实现和测试自动缩放时，您将主要使用以下内容

  我们在这里学到的操作： -- 将文件从上传存储桶复制到处理存储桶以开始处理它们（任务 K） -- 从队列中清除（清空）消息（使用 SQS 控制台中的“清除”按钮 -

  任务 H(1)) -- 检查 EC2 实例上的工作日志（任务 I/K）

-- Stopping or starting the workfreq worker service if necessary on an EC2 instance (Task I)
-- Running the run_worker.sh script on the Dev Instance for initial testing (Task H)

NOTE: The infrastructure configuration we have performed is functional, but not necessarily optimal or best practice...

IMPORTANT NOTES

- When you have finished a coursework session, ensure that any EC2 instances are stopped to minimise cost.
  NOTE 1: The approximate cost for a running EC2 t2.micro instance is about 2 cents (US) per hour, but it adds up if never stopped.
  NOTE 2: You do not pay for stopped EC2 instances, but you still pay for their EBS storage volumes, however this is a fraction of the EC2 cost.
- When you restart an EC2 instance, the free Public IP changes, so if you are accessing SSH via the IP address only, you will need to copy the new one.

STRONG Recommendation: Create an AMI Backup before ending this session!!

- Create an AMI image from the running EC2 instance (Instances > select wordfreq-dev > Actions > 'Image and templates' > Create image).
- If you lose your configured EC2 instance, check with the instructors on how to retrieve your configuration from the AMI, or rebuild a new EC2 instance as above.
- EBS Snapshots can also be used to store incremental backups of an EBS disk volume used by an instance.

SUPPORT

For any general or technical issues with this setup, please start a new post on the BlackBoard Coursework Discussion Forum
Alternatively, for one-to-one support, please book an Office Hours Session with the instructor or a TA.

-- 如有必要，在 EC2 实例上停止或启动 workfreq 工作线程服务（任务 I）
-- 在开发实例上运行 run_worker.sh 脚本进行初始测试（任务 H）

注意：我们执行的基础设施配置是有效的，但不一定是最佳或最佳实践......

重要注意事项

- 当您完成课程作业后，请确保停止所有 EC2 实例以最大限度地降低成本。

  注 1：运行的 EC2 t2.micro 实例的大约成本约为每小时 2 美分，但如果从未停止，成本会不断增加。

  注意 2：您无需为已停止的 EC2 实例付费，但仍需为其 EBS 存储卷付费，但这只是 EC2 成本的一小部分。

- 当您重新启动 EC2 实例时，免费公共 IP 会发生变化，因此如果您通过 SSH 访问
  仅 IP 地址，您需要复制新的 IP 地址。

强烈建议：在结束本次会话之前创建 AMI 备份！

- 从正在运行的 EC2 实例创建 AMI 映像（实例 > 选择 wordfreq-dev > 操作 > “映像和模板” > 创建映像）。
- 如果您丢失了配置的 EC2 实例，请向讲师咨询如何从 AMI 检索您的配置，或按照上述方式重建新的 EC2 实例。 EBS 快照还可用于存储实例使用的 EBS 磁盘卷的增量备份。
-

支持

对于此设置的任何一般或技术问题，请在 BlackBoard 上开始新帖子
课程作业讨论论坛 或者，如需一对一支持，请与讲师或助教预订办公时间会议。